Exploratory Data Analysis with Coffee Sales Data and MNIST Data Clustering

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1. Abstract

This project involves exploratory data analysis on a real-world coffee sales dataset and unsupervised machine learning on the MNIST handwritten digits dataset. The coffee sales dataset consists of 3,547 transaction records from a coffee shop spanning the years 2024 and 2025, containing variables such as coffee type, time of day, day of the week, and transaction amount. Descriptive statistics and aggregation techniques have been employed to uncover sales patterns, with careful distinction between truly numerical variables and ordinal variables such as hour of day, month sort, and weekday sort, for which only order-based statistics have been reported. The analysis reveals that Latte and Americano with Milk are the highest revenue-generating coffee types, while sales remain relatively consistent across morning, afternoon, and night periods. In the machine learning component, the sklearn digits dataset, a compact version of the MNIST dataset containing 1,797 handwritten digit images from 0 to 9 has been used, where each image is represented as a flattened vector of 64 pixel values from an 8x8 grid. K-Means clustering with 10 clusters has been applied to the image features without the use of any class labels, making it a purely unsupervised approach. Since K-Means assigns arbitrary cluster identifiers, each cluster has been mapped to the most frequently occurring true digit label within it. The performance has been evaluated using the macro-averaged F1 score, which is found to be approximately 0.79, and a confusion matrix has further been used to examine misclassification patterns across the ten digit classes.

2. Introduction

Data analysis and machine learning have become indispensable tools in modern science and industry. The ability to extract patterns from raw data and build intelligent systems has transformed fields ranging from retail and healthcare to finance and technology. This project, undertaken as part of the IDEAS Summer Internship Program 2026 at ISI Kolkata, provides hands-on experience with two fundamental data science tasks — exploratory data analysis on a real-world coffee sales dataset and unsupervised machine learning on the MNIST handwritten digits dataset.

Relevance

Understanding sales patterns is critical for business decision-making in the retail and food service industry. Exploratory data analysis is the first and most important step in any data science workflow, helping the analyst understand data structure, quality, and key characteristics before applying advanced techniques. Clustering is widely used in real-world applications such as image recognition, customer segmentation, and anomaly detection. The ability to group unlabelled data into meaningful clusters without any prior supervision makes K-Means one of the most practically relevant algorithms in modern machine learning.

Technology Involved

The project has been implemented in Python using Google Colab. The key libraries used are pandas for data manipulation, NumPy for numerical computing, and scikit-learn for machine learning tasks including K-Means clustering, F1 score computation, and confusion matrix generation.

Background Material Survey

Exploratory data analysis is a methodology introduced by John Tukey (1977) that emphasises statistical and graphical tools for summarising data. A key statistical distinction observed in this project is between numerical variables, for which all descriptive statistics are valid, and ordinal variables, for which only order-based measures such as quartiles are meaningful. K-Means clustering, first proposed by Stuart Lloyd (1957) and popularised by MacQueen (1967), partitions data into K clusters by minimising within-cluster distances. The sklearn digits dataset is a compact version of the MNIST dataset introduced by LeCun et al. (1998), containing 1,797 handwritten digit images at 8x8 pixel resolution.

Procedure Used

The project follows the ten questions provided in the project notebook. The coffee sales data was loaded, inspected for quality, and analysed using descriptive statistics and groupby aggregations. A synthetic dataset was also generated to practise date and random number operations. For the machine learning component, K-Means clustering was applied to the digit image features, cluster labels were mapped to true digit labels, and performance was evaluated using the macro-averaged F1 score and confusion matrix.

Purpose

The primary purpose of this project is to apply and demonstrate the skills acquired during the internship, provide hands-on experience with real-world data, and develop the ability to independently analyse data and evaluate machine learning models using Python.

Topics Covered During the Internship

Session 1 — Introduction and Orientation: Orientation by the IDEAS TIH team covering a welcome address, a session on Thinking with AI, internship framework overview, project path identification, and a quiz and self-assessment framework.

Session 2 — Python Basics 1: Variables, data types, type casting, arithmetic operations, string operations, conditional statements, for loops, and lists.

Session 3 — Python Basics 2: Advanced string operations, while loops, functions, recursion, lambda functions, dictionaries, sets, and tuples.

Session 4 — Python Basics 3 (Object Oriented Programming): Classes, objects, constructors, self keyword, class and instance attributes, static methods, abstraction, and inheritance.

Session 5 — Python Basics 4 (NumPy and Pandas): NumPy arrays, array operations, linear algebra functions, and Pandas DataFrames, CSV operations, indexing, filtering, sorting, and descriptive statistics.

Session 6 — Machine Learning Overview: Introduction to regression, classification, clustering, and dimensionality reduction.

Session 7 — Machine Learning 1: Linear regression, logistic regression, and introduction to tensors.

Session 8 — Machine Learning 2: Supervised vs unsupervised learning, K-Means clustering, and DBSCAN.

Session 9 — Large Language Models: Tokenisation, word embeddings, cosine similarity, pre-training, supervised fine tuning, RLHF, and the history and architecture of language models.

Session 10 — Communication Skills for Data Scientists: Shannon-Weaver's and Berlo's SMCR models, Pyramid Principle, formal and informal communication, and effective communication strategies for data scientists.

3. Project Objective

The objectives of this project are as follows:

- To load, inspect and explore the coffee sales dataset by examining its shape, checking for missing values and duplicate records, and understanding the nature and measurement scale of each variable.
- To compute appropriate descriptive statistics for the dataset and perform necessary data type conversions such as converting the Date column from object to datetime format and extracting relevant features from it.
- To perform data aggregation and grouping operations to extract meaningful business insights such as average sales per year, maximum sales by month, total revenue by coffee type, and average sales by time of day.
- To generate a synthetic dataset using Python's date and random number utilities to demonstrate data creation independently of a real dataset.
- To apply K-Means clustering on the MNIST digits dataset and evaluate its performance using the macro-averaged F1 score and confusion matrix.

4. Methodology

4.1 Data Collection

The coffee sales dataset was provided as part of the project by the project guide. It was downloaded from the Google Drive link provided in the project notebook and uploaded into the Google Colab environment using the files.upload() utility from google.colab. The dataset was then loaded into a pandas DataFrame named coffee_data using pd.read_csv(). The MNIST digits dataset was loaded directly from the scikit-learn library using load_digits(), requiring no external download.

4.2 Data Inspection and Cleaning

The dataset was inspected for its basic properties using .shape to determine the number of rows and columns, .duplicated().sum() to check for duplicate records, and .isnull().sum().sum() to check for missing values. The dataset was found to contain 3,547 rows and 11 columns with no duplicate records and no missing values, indicating that no data cleaning was required.

4.3 Data Type Conversion and Feature Extraction

The Date column was originally stored as plain text (object type). It was converted to datetime format using pd.to_datetime() with the format parameter explicitly set to '%Y-%m-%d' to avoid parsing errors. Two new columns — Month and Year — were then extracted from the converted Date column using .dt.month and .dt.year respectively.

4.4 Descriptive Statistics

Descriptive statistics were computed with careful attention to the measurement scale of each variable. For the money column, which is a ratio-scale numerical variable, all standard statistics including mean, standard deviation, and quartiles were computed using .describe(). For the ordinal variables hour_of_day, Weekdaysort, and Monthsort, only order-based statistics — count, minimum, maximum, and quartiles — were reported by filtering the .describe() output using .loc[['count', 'min', '25%', '50%', '75%', 'max']], as mean and standard deviation are not meaningful for ordinal data.

4.5 Data Aggregation and Analysis

Several aggregation tasks were performed using the groupby() method in pandas. The data was grouped by Year to compute average sales, by Year and Monthsort to find maximum sales in calendar order, and by coffee_name to calculate total revenue. Unique coffee types were identified using nunique() and unique(). Average sales by Time_of_Day were also computed. The display() function was used throughout to render results as properly formatted pandas tables.

4.6 Synthetic Data Generation

A synthetic DataFrame of 100 rows was created independently of the coffee sales data. pd.date_range() was used to generate 100 consecutive daily dates starting from 2023-01-01, and np.random.randn() was used to generate 100 random values from a standard normal distribution. np.random.seed(42) was set to ensure reproducibility.

4.7 K-Means Clustering on MNIST Data

The sklearn digits dataset was loaded using load_digits() and separated into features X (shape 1797×64) and true labels y. K-Means clustering was applied using KMeans from sklearn.cluster with n_clusters=10, random_state=42, and n_init=10. The algorithm was fitted on X without using any label information, making it a purely unsupervised approach. Since K-Means assigns arbitrary cluster numbers, each cluster was mapped to the most frequently occurring true digit label within it using np.bincount().argmax(). The mapped labels were then compared against the true labels y to evaluate performance.

4.8 Model Evaluation

The performance of the clustering was evaluated using two metrics. The macro-averaged F1 score was computed using f1_score() from sklearn.metrics with average='macro', which calculates the F1 score for each digit class separately and takes their average, giving equal weight to all classes. A 10×10 confusion matrix was generated using confusion_matrix() from sklearn.metrics and displayed as a labeled pandas DataFrame to examine misclassification patterns across the ten digit classes. Since K-Means is an unsupervised algorithm, there is no train-test split involved in this project.

Flowchart:

Load Data → Inspect Data → Data Type Conversion → Descriptive Statistics → Aggregation → Synthetic Data → Load MNIST → K-Means Clustering → Label Mapping → F1 Score → Confusion Matrix

5. Data Analysis and Results

5.1 Dataset Overview

The coffee sales dataset contains 3,547 rows and 11 columns. No missing values or duplicate rows were found. The columns include hour_of_day, cash_type, money, coffee_name, Time_of_Day, Weekday, Month_name, Weekdaysort, Monthsort, Date, and Time. After data type conversion, two additional columns — Month and Year — were added, bringing the total to 13 columns.

5.2 Descriptive Statistics

For the money column, the mean transaction value is Rs. 31.65 with a standard deviation of Rs. 4.88, indicating that transaction values are fairly consistent with low spread. The minimum transaction value is Rs. 18.12 and the maximum is Rs. 38.70. For the ordinal variables, the hour_of_day ranges from 6 to 22 with a median of 14, Weekdaysort ranges from 1 to 7 covering all days of the week, and Monthsort ranges from 1 to 12 covering April to December of 2024 and January to March of 2025.

5.3 Sales Aggregation Results

Average sales per year were Rs. 31.74 in 2024 and Rs. 31.39 in 2025, indicating stable pricing and sales patterns across both years. Maximum sales by month showed that March and April 2024 had the highest maximum transaction value of Rs. 38.70, while values dropped to Rs. 35.76 from September 2024 onwards. Total revenue by coffee type revealed that Latte was the highest earner at Rs. 26,875.30, followed by Americano with Milk at Rs. 24,751.12, while Espresso had the lowest revenue at Rs. 2,690.28. There are 8 unique coffee types in the dataset. Average sales by time of day were fairly consistent across morning, afternoon and night, indicating no strong peak sales period.

5.4 Synthetic Dataset

A synthetic DataFrame of 100 rows was successfully created with date and value columns. Dates range from 2023-01-01 to 2023-04-10 and values follow a standard normal distribution ranging roughly between -3 and +3.

5.5 MNIST Dataset

The sklearn digits dataset was successfully loaded with features X of shape (1797, 64) and labels y of shape (1797,). The dataset contains 10 digit classes from 0 to 9 with roughly 180 images per class, making it a fairly balanced dataset.

5.6 K-Means Clustering Results

K-Means clustering with 10 clusters was successfully fitted on the digit image features. The cluster-to-digit mapping revealed that each cluster was dominated by a specific digit — for example cluster 0 mapped to digit 2, cluster 1 to digit 0, and so on. The macro-averaged F1 score was found to be **0.7895**, indicating that the algorithm correctly identified approximately 79% of the digit images without any supervision.

5.7 Confusion Matrix Analysis

The 10×10 confusion matrix revealed the following key findings:

- Digits 0 and 6 were identified almost perfectly with 177 correct predictions each.
- Digit 7 also performed well with 170 correct predictions.
- Digit 1 had the most misclassifications — 99 out of 182 images were incorrectly mapped to digit 8, possibly due to the low 8x8 pixel resolution making fine differences harder to capture.
- Digit 8 showed confusion with digit 9, with 48 misclassifications.
- Digit 5 had 41 images confused with digit 9.

6. Conclusion

The exploratory data analysis on the coffee sales dataset led to several meaningful conclusions. Latte and Americano with Milk are the highest revenue-generating coffee types with total revenues of Rs. 26,875.30 and Rs. 24,751.12 respectively, while Espresso had the lowest revenue at Rs. 2,690.28. Average sales remained stable across 2024 (Rs. 31.74) and 2025 (Rs. 31.39), suggesting consistent pricing and demand. The absence of any strong peak period across morning, afternoon and night indicates that the coffee shop maintains consistent footfall throughout the day.

An important conclusion drawn from the descriptive statistics is that mean and standard deviation are not meaningful for ordinal variables such as hour_of_day, Weekdaysort and Monthsort, and only order-based measures such as quartiles should be reported for such variables.

The K-Means clustering on the MNIST digits dataset achieved a macro-averaged F1 score of 0.7895, demonstrating that the algorithm was able to correctly identify approximately 79% of the digit images without any label information. The confusion matrix confirmed that digits 0 and 6 were identified almost perfectly with 177 correct predictions each, while digit 1 had the most misclassifications with 99 out of 182 images being incorrectly mapped to digit 8, likely due to the low 8x8 pixel resolution of the dataset making fine structural differences harder to capture.

As a recommendation for future work, applying dimensionality reduction techniques such as PCA prior to clustering may improve performance by reducing noise in the pixel features. Additionally, comparing K-Means with other clustering algorithms such as DBSCAN and hierarchical clustering on the same dataset would provide a more comprehensive evaluation of unsupervised learning approaches on image data.

7. APPENDICES

Appendix A: References

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- pandas documentation. Available at: https://pandas.pydata.org/docs/
- scikit-learn documentation. Available at: https://scikit-learn.org/stable/documentation.html
- NumPy documentation. Available at: https://numpy.org/doc/

Appendix B: GitHub Link for Code

GitHub Repository Link:

https://github.com/rudrashmibiswas/Rudrashmi_Biswas_IDEASTIH_2026_